Feature Specification Document

Team Members: Yunhao Chen, Hizo Chen, Fan Yin, Elizabeth

1. Context of Kingdom Rush 5

Game Name: Kingdom Rush 5: Alliance
Game Type: Tower Defense
Release Date: July 15, 2024
Developer: Ironhide Game Studio

1.1 Introduce

Kingdom Rush 5 is a classic tower defense game where players place towers at specific sites along the paths to destroy waves of enemies appearing in predetermined intervals. The goal is to defeat all enemy waves before they reach the end of the road (core). Player win the game if fewer than a certain number of enemies reach the **core** after all waves are defeated.



Figure 1: Placing defense towers



Figure 2: Defeat enemies



Figure 3: Enemies reach the core



Figure 4: Enemies pass core deduct health



Figure 5: Game over if health are depleted



Figure 6: Victory after having defeated all enemies

Players are scored based on remaining health of core after completing a level.

1.2 Hero System

Kingdom Rush 5 introduces units that engage enemies along the path, generated through towers or skills.



Figure 7: Barracks units (engage enemies)



Figure 8: Reinforcement units (engage enemies)

Heroes are unique, controllable units with stronger combat abilities.



Figure 9: Hero units (engage enemies)

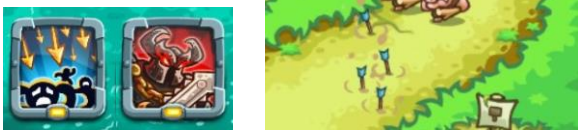


Figure 10: Hero skill using

As levels progress, new towers and heroes will be unlocked. Heroes gain experience by defeating enemies. After reaching the full experience of the current level, the hero will level up and become more powerful. Heroes can level up to a maximum of level 10.

In the **Hero Interface**, players can view detailed information about each **hero** by selecting their avatar.

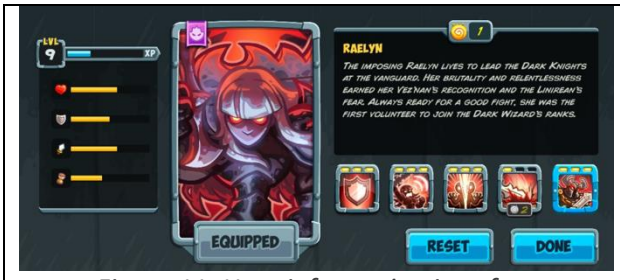


Figure 11: Hero information interface

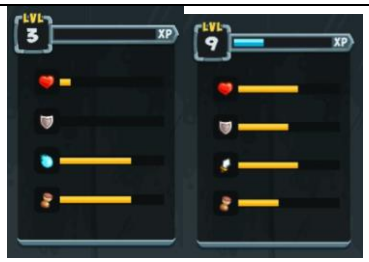


Figure 12: Hero levels info from different heroes

When heroes level up, they gain **SkillPoints**, which can be used to upgrade skills. Skill upgrades are reversible. Players is allowed to reset and redistribute skillpoints.



Figure 13: using skillpoints to upgrade

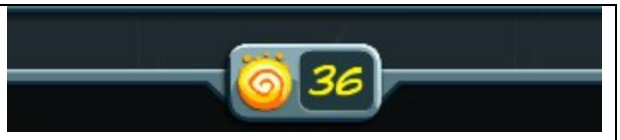


Figure 14: Skillpoints

1.3 Level System

From the main menu, players can select **Main Levels** and access information interfaces for **Heroes**, **Towers**, and the **Upgrades**. Completing a level unlocks the next level.



Figure 15: Main menu



Figure 16: Level unlock

1.4 Equipment

1.4.1 Hero Equipment

In the **Hero Interface**, there are 2 hero equipment slots for heroes, who will applied in battle.

To change equipped heroes, players click on the desired **hero's** avatar, press the equip button, and select the slot to replace the existing **hero**.

1.4.2 Tower Equipment

In the **Tower Interface**, there are equipment slots for towers will applied in battle. (Maximum 5 towers)
The process for changing towers is the same as for heroes.

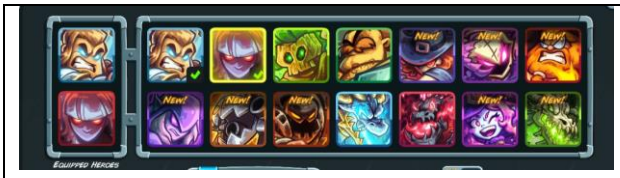


Figure 17: hero equipment slots



Figure 18: tower equipment slots

1.5 Game Story

At the beginning or end of level, the game presents story as comic-style to narrate the story.

Main Plot:
Following ***Kingdom Rush: Vengeance***, Linirea falls under **Vez'nan's** rule. Meanwhile, a rebellion is brewing in the southern forests. The kingdom faces a threat from interdimensional beings called the "Warlords." **Vez'nan's** dark army forms an uneasy alliance with **Linirea's** rebels to counter this threat. The heroes of Linirea—including elves, orcs, and dark knights—unite their strength to protect their world.



Figure 19: comic-style of game story

2. Received Goal

Each hero will have corresponding unique Branch Levels catering to various **Player Persona Types**:

“Commander”	**“Hero”**
Enjoy strategic planning and precise execution.	Interested in character upgrades and growth.
Typical players of Kingdom Rush and similar strategy games.	Value character stories and lore.
Feedback often mentions the need for more levels to enhance the depth of experience.	May limited strategic abilities.

Objectives:

1. Provide players with deeper engagement with game units.
2. Introduce easier-to-follow guidance and rewarding mechanisms for less strategy-oriented players.
3. Add varied challenges to give “Commander” players more gameplay options.
4. Enhance hero personalities in the story, strengthening emotional connections for “Hero” players.

3. Systems and Mechanics

3.1 Branch Level Selection

3.1.1 How to Select Branch Levels

Open the Character Interface to access **Branch Levels**.



Figure 20: Heros info Interface with Branch Level added

Each character has **1–3** Branch Levels that displayed on the side of interface.



Figure 21: Level icons for heroes with 3 branch level



Figure 22: Level icons for heroes with 2 branch level



Figure 23: Level icons for heroes with 1 branch level

If a level rewards skill points, a skillpoint icon will be displayed in the top-right corner of the level icon.
if two heroes share one Branch Level, the level icon will show the avatar of the other hero.



Figure 24: Skillpoint icon



Figure 25: Related hero avatar

When players click on the Branch Levels, they will enter the basic information interface for that branch level.

3.1.2 Level Unlocking

Unlocked and locked levels will be displayed in different forms.



Figure 26: Unlocked level



Figure 27: Locked level

When mouse pointer hovers over an unlocked level, it will display the unlocking conditions.



Figure 28: Branch Level unlocking conditions

3.1.3 Tutorial Introduction

After completing Mainline Level 2, the hero icon in the main menu will be highlighted, prompting the player to open the character interface.

The player will be informed that they can unlock the Vesper's (the first unlocked hero) first Branch Level and will be forced to open the information interface of this level. Afterward, the player will be free to proceed with their actions.



Figure 29: Highlighted the Heroes UI button



Figure 30: Forced to open Vesper's first Branch Level



Figure 31: Forcing the player to view the information interface

3.2 Branch Level Flow

Each Branch Level will have a tutorial theme, details are in the attached *Excel Branch Level: Sheet 2 Level Content*. Using Raelyn's first Branch Level as an example, we will demonstrate how Branch Levels are designed and played.

- More details about other Branch Level see *Excel Branch Levels: Sheet 1 Level Unlock.*

3.2.1 Information Interface

After selecting a Branch Level, players will enter the information interface, where a story summary and tutorial theme will be displayed.



Figure 32: Realyn's first Branch Level information interface

Click the "Battle!" button to enter the hero equipment checking interface. (Note that this is different from the mainline levels, which directly lead to the game). Click the "Next" button arrow to enter the tower equipment interface.



Figure 33: Hero equipment checking interface



Figure 34: Tower equipment checking interface

Each Branch Level will have specific equipment requirements, requiring players to bring certain heroes or towers. If the player does not meet the equipment requirements for the level, the required hero or tower will automatically be equipped in the player's first equipment slot. Players can freely change the equipment, but they can only start the battle if their equipment meets the requirements of Branch Levels.The "Battle!" button will light up when the equipment satisfies the requirements, allowing players to click it.

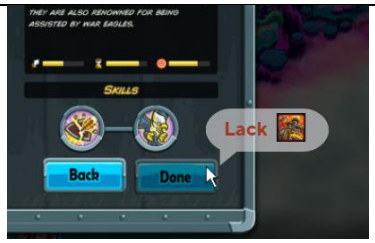


*Figure 35: **Equipment does not meet the requirements***

3.2.2 Story Segment 1

If the branch level's story requires setting up the plot or character development, a comic will be shown at the beginning of the level.

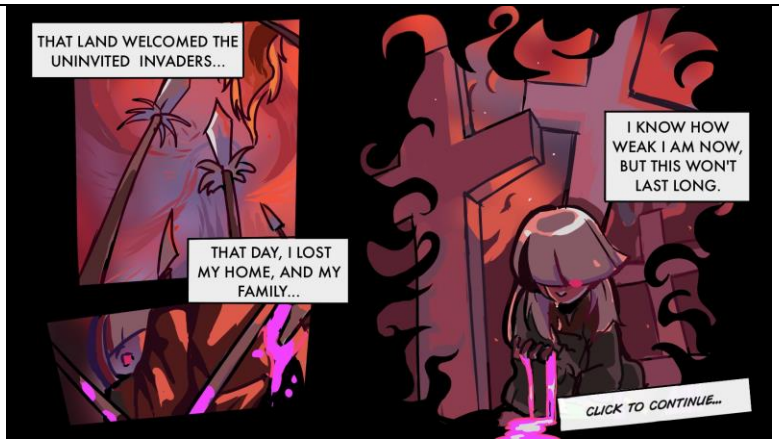


Figure 37: Raelyn's first Branch Level story

3.2.2 Preparation Phase

In the Branch Level, some towers may already builded on the field and cannot be sold during the levels.
In Raelyn's first Branch Level, the tower initial setup is: **1 Archer Tower** (Locked)

After entering the level, a dialogue box will appear above the character, containing hints about the level. The text will appear in a typewriter style. Once all the text is displayed, the player can click mouse button to proceed to the next line of dialogue. If the player click mouse button during the typing process, the current line of dialogue will instantly complete.

Once the player is ready, they will click the wave spawn button to release the first wave of enemies.



Figure 38: Archer Tower that selling is prohibited



Figure 39: Character speech bubble

3.2.3 Phase 1 (Waves 1-2)

Phase Objective: Allow players to familiarize with hero skills without applying too much pressure.
Wave1 & Wave2: Hog Invader and **Tusked Brawler**.

3.2.4 Phase 2 (Waves 3-4)

Phase Objective: Guide players to understand the hero's strengths and weaknesses.
Wave 3: Bear Vanguard (Single-target elite enemy, suitable for Raelyn)
Wave 4: Cutthroat Rat (Fast, spawn in groups, suitable for ranged heroes instead of Raelin)

3.2.5 Phase 3 (Waves 5-6)

Phase Objective: Provide strategic guidance, including specific strategy combinations or familiarization with tower usage.
Wave5 & Wave6: Hog Invader (Low health, spawn in groups, suitable for ranged heroes instead of Raelin)
Guidance: Guide the player to use the Triple-Barrel Cannon to clear groups of enemies quickly with the tower's AOE (Area of Effect).

3.2.6 Phase 4 (Wave 7)

Phase Objective: Help players assimilate all the skills learned so far by mixing previously encountered enemies.
Wave 7: Hog Invader, Tusked Brawler, Cutthroat Rat, Bear Vanguard.

- *More Details about Wave instantiation see Excel Branch Levels: Sheet 3 Wave Insatntiation.*

3.2.7 Victory and Star Rating Determination

Victory or defeat is determined based on the remaining health. The number of rating stars is based on the remaining health as well.

3.2.8 Story Segment 2

If the level includes planned story developments or conclusions, a comic will be displayed at the end of the level.

3.3 Branch Level Rewards

3.3.1 Level Unlock

When a hero's Branch Level is completed, players will receive a level unlock reward.
In the original game, heroes could level up to 10 with no restriction. In the modified system, Level 9 and Level 10 are locked, meaning heroes can only reach Level 8 before they get the level unlock reward Even if a hero is at Level 8 with full experience, they cannot level up to Level 9. However, heroes can continue accumulating experience, and once the level is unlocked, they can upgrade.
When Level 9 or Level 10 is locked, players at Level 8 or Level 9 will see an icon indicating a level-up restriction on the hero's experience bar. Hovering over the icon will display an explanation.

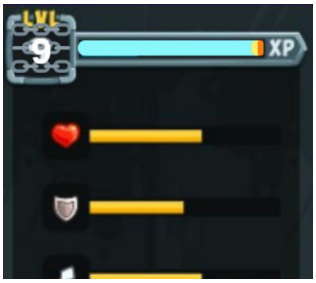


Figure 39: Locked Level



Figure 40: Level unlock condition explanation

The level unlocks rewards have some differences depending on the number of Branch Levels the hero has：

The number of Branch Levels the hero has：	The level unlocks rewards
1	unlock Level 9 in Branch Level 2 unlock Level 10 in Branch Level 3
2	unlock Level 9 in Branch Level 1 unlock Level 10 in Branch Level 2
3	unlock both Level 9 and Level 10 in Branch Level 1.

3.3.2 Skill Point Rewards

In the original system, characters received 4 skill points per level. In the updated system, the skill points earned for leveling up from Level 8 to 9 and from Level 9 to 10 are reduced to 2 skill points each.
The 2 skill points that are reduced for reaching Level 9 or Level 10 will be granted when those levels are unlocked (as described in Section 3.3.1).
Example:

- A player completes Realyn's Branch Level 2 at Level 4 (Realyn has 3 Branch Levels in total), unlocking Realyn's Level 9 and receiving 2 skillpoints.

- Similarly, when a player completes Nero's Branch Level 2 at Level 8 (with Nero at full experience), they unlock Nero's Level 9 and receive 2 skillpoints. Since the player is already at full experience at Level 8, Nero immediately levels up to Level 9 and receives 2 skillpoints.

3.4 Story System

Each character's Branch Levels serve a narrative purpose. The narrative structure have some differences depending on the number of Branch Levels (Section 3.1) the hero has:

The narrative structure of different branch level

The number of Branch Levels the hero has：	first branch level	second branch level	third branch level	
1	Setup	confrontation	resolution	
2	Setup + Half of confrontation + resolution	Half of confrontation + resolution	/	
3	Setup + confrontation + resolution*	/	/	

**tips: Setup + Half of confrontation +resolution in Story Segment 1, Half of confrontation +resolution in Story Segment 2*